

2.6.5) Gamma DistributionDef. 2.13

$$\Gamma(k) = \int_0^{\infty} t^{k-1} e^{-t} dt \quad (k > 0).$$

(e.g.)

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = [-e^{-t}]_0^{\infty} = 1.$$

Properties

$$(i) \quad \Gamma(k) = (k-1) \Gamma(k-1) \quad (k > 1).$$

$$(ii) \quad \Gamma(n) = (n-1)! \quad (n \in \mathbb{N}).$$

$$(iii) \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

pf). (i). Let $u = x^{k-1}$
 Let $dv = e^{-x} dx \rightarrow \begin{aligned} du &= (k-1) x^{k-2} dx. \\ v &= -e^{-x}. \end{aligned}$

$$\begin{aligned} \Rightarrow \int_0^{\infty} x^{k-1} e^{-x} dx &= [x^{k-1} e^{-x}]_0^{\infty} - \int_0^{\infty} (-e^{-x}) (k-1) x^{k-2} dx. \\ &= (k-1) \int_0^{\infty} x^{k-2} e^{-x} dx. \\ &= (k-1) \Gamma(k-1). \end{aligned}$$

②.

• pdf of gamma r.v. X .

For $\theta, k > 0$;

$$f(x; k, \theta) = \frac{1}{\theta^k \Gamma(k)} x^{k-1} e^{-\frac{x}{\theta}}, \quad x > 0.$$

Denote as $X \sim \text{GAM}(k, \theta)$.

$$E[X] = \frac{1}{\theta^k \Gamma(k)} \int_0^{\infty} x^{(1+k)-1} e^{-\left(\frac{x}{\theta}\right)} dx.$$

$$= \frac{\theta^{1+k} \Gamma(k+1)}{\theta^k \Gamma(k)} \int_0^{\infty} \frac{1}{\theta^{1+k} \Gamma(k+1)} x^{(1+k)-1} e^{-\left(\frac{x}{\theta}\right)} dx.$$

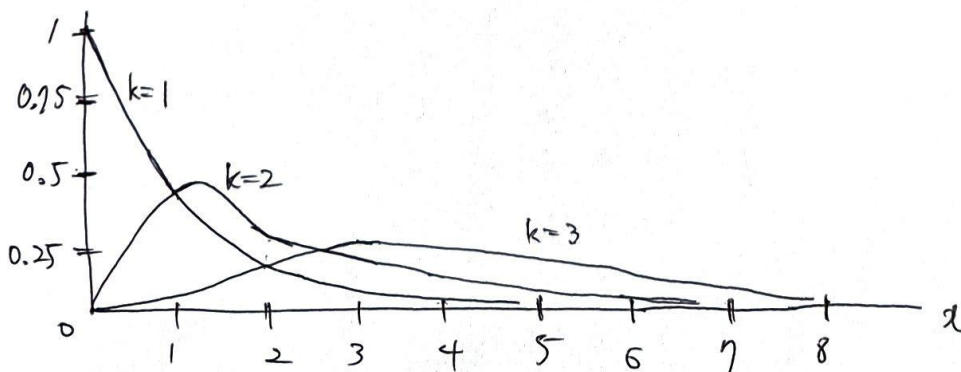
$$= \frac{\theta^{1+k} \Gamma(k+1)}{\theta^k \Gamma(k)} = k\theta.$$

Likewise, $E[X^2] = k(1+k)\theta^2$.

$$\rightarrow \text{Var}(X) = k(k+1)\theta^2 - (k\theta)^2 = k\theta^2.$$

k = shape parameter.

If $k=1$, then it is an exponential dist. of parameter θ .



$$X \sim \text{GAM}(1, k).$$

2.6.6 Beta Distribution $(0, 1)$ interval.Def. 2.14

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx. \quad (a > 0, b > 0).$$

• pdf of beta r.v. X :

$$f(x; a, b) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1} \quad (0 < x < 1).$$

" $X \sim \text{Beta}(a, b)$."For $k > -a$,

$$E[X^k] = \frac{1}{B(a, b)} \int_0^1 x^{k+a-1} (1-x)^{b-1} dx.$$

$$= \frac{B(a+k, b)}{B(a, b)} = \frac{\Gamma(a+k)\Gamma(b)}{\Gamma(a+b+k)} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}$$

$$= \frac{\Gamma(a+k)\Gamma(a+b)}{\Gamma(a)\Gamma(a+b+k)}.$$

④.

$$E[X] = \frac{\Gamma(a+1) \Gamma(a+b)}{\Gamma(a) \Gamma(a+b+1)} = \frac{a}{a+b}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

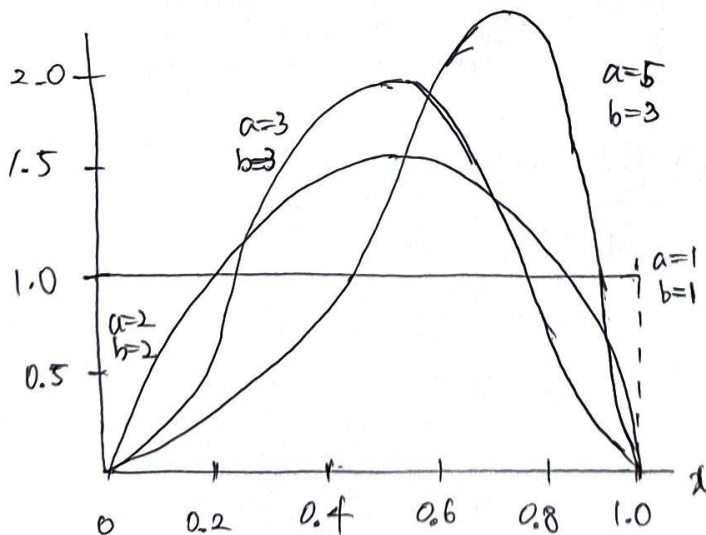
$$= \frac{\Gamma(a+2) \Gamma(a+b)}{\Gamma(a) \Gamma(a+b+2)} - \left(\frac{a}{a+b}\right)^2$$

$$= \frac{(a+1)a}{(a+b+1)(a+b)} - \left(\frac{a}{a+b}\right)^2 = \frac{ab}{(a+b+1)(a+b)^2}$$

Note. $B(a,b) = \frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)}$

If $a=b=1$, $B(1,1) = U(0,1)$.

Beta dist. $\rightarrow$ prob. model for any finite interval.



$$X \sim \text{BETA}(a,b).$$